Exercises

1. Explain the structure of a basic autoencoder. What are the roles of: encoder, latent space, and decoder. Why is the bottleneck important?
2. Autoencoders are trained using a reconstruction loss. What is the reconstruction loss? Compare **MSE loss vs Binary Cross-Entropy loss** in autoencoders. When would you choose one over the other?
3. Implement a fully connected autoencoder for MNIST: **Input**: 28×28 images**, Latent dimension**: 32, **Loss**: MSE
4. What happens if the latent dimension is too large? Why can an autoencoder "memorize" data? How does this differ from generalization in supervised learning?
5. Modify a basic autoencoder to become a **denoising autoencoder**: Add Gaussian noise to input images, Train model to reconstruct clean images, Evaluate reconstruction quality
6. Explain how VAEs differ from traditional autoencoders.
7. Why is KL divergence used in VAEs? What does it enforce on the latent space? What happens if KL loss is removed?
8. Implement the reparameterization trick. Given: **mean (μ)**, **log variance ($\log \sigma^2$). Sample $z = \mu + \sigma * \varepsilon$.** Explain why this is needed for backpropagation.
9. Build a full VAE with: Encoder outputs: μ and $\log \sigma^2$, Decoder reconstructs images, and Loss = reconstruction loss + KL divergence. Train on MNIST and generate new samples from latent space.
10. After training a VAE: 1) Reduce latent space to 2D (if not already), 2) Plot latent space distribution, 3) Interpolate between two points and visualize decoded outputs.